CS 306: Introduction

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https://www.iiti.ac.in/people/~ayanm/

Outline

- Course Information
- Data Communications
- Networks
- Network Types
- Basics of Network Models

Course Information

- Course webpage

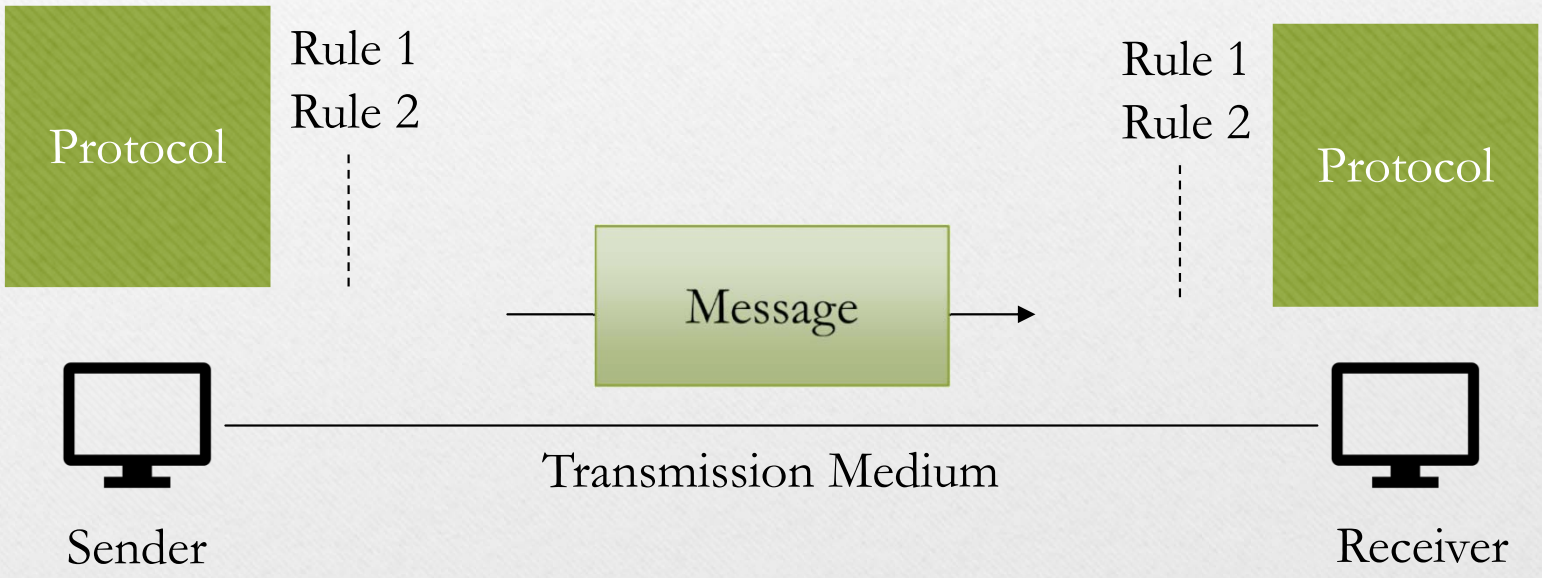
  https://www.iiti.ac.in/people/~ayanm/Courses/Spring_2022/CS306/

- Course evaluation
  - Attendance: 10%
  - Assignments + Class Test: 20%
  - Mid Sem Exam: 25%
  - Final Exam: 45%

Data Communications

- When we communicate, we are sharing information. This sharing can be local or remote. The term telecommunication, which includes telephony, telegraph, and television, means communication at a distance.
- Data communications are the exchange of data between two devices via some form of transmission media.

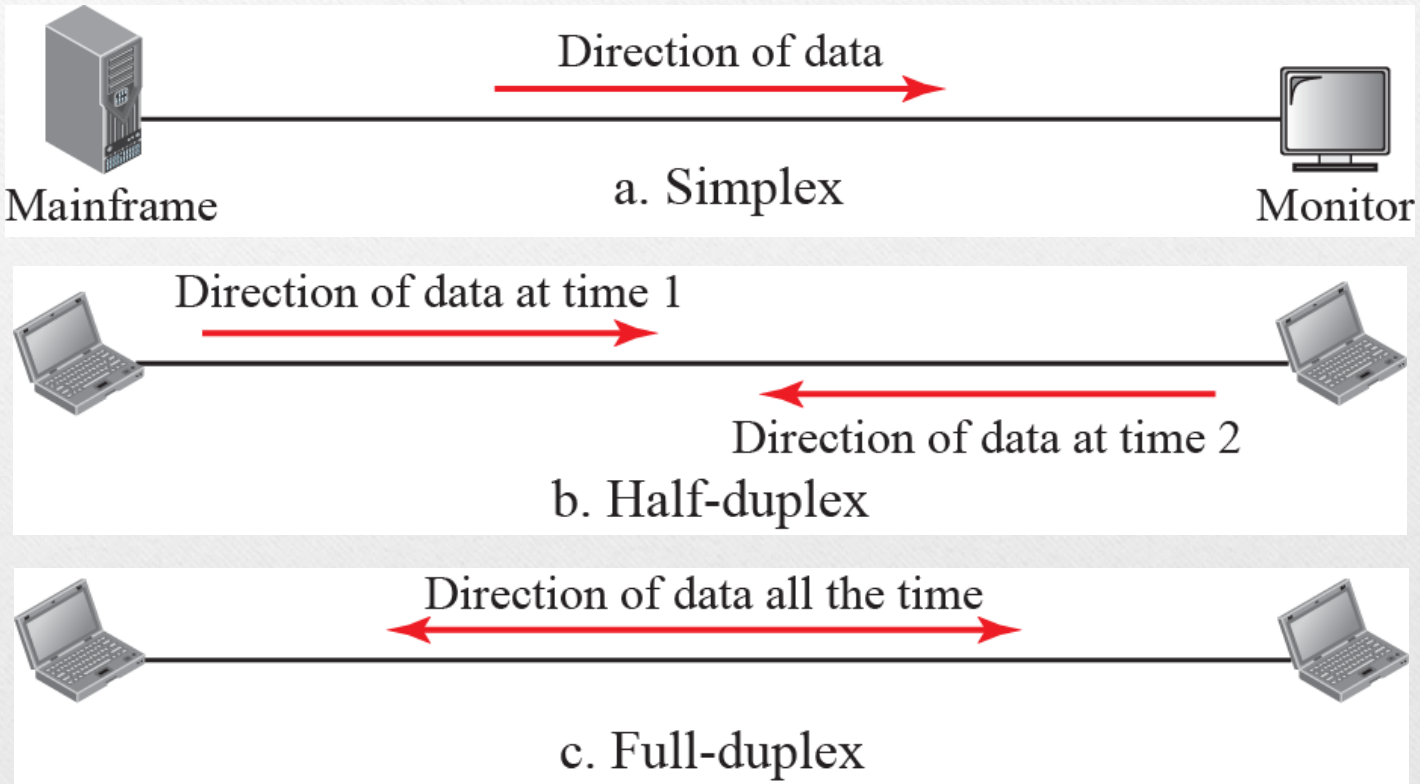
Components


Protocol
Rule 1
Rule 2
Message
Rule 1
Rule 2
Protocol
Transmission Medium
Sender
Receiver

Data Representation

- Information today comes in different forms such as text, numbers, images, audio, and video

Data Flow


Direction of data
Mainframe
a. Simplex
Monitor
Direction of data at time 1
Direction of data at time 2
b. Half-duplex
Direction of data all the time
c. Full-duplex

Networks

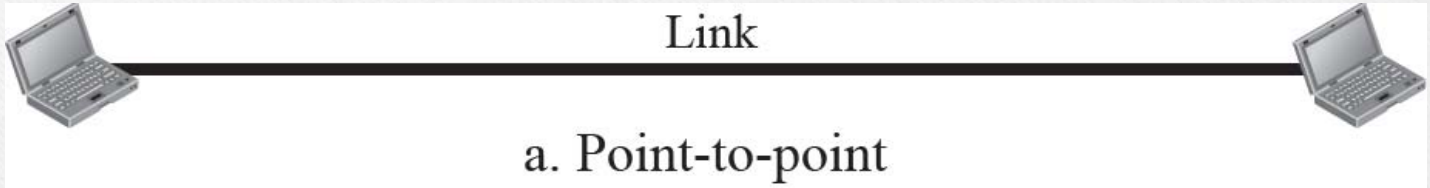
- A network is the interconnection of a set of devices capable of communication. In this definition, a device can be a host such as a large computer, desktop, laptop, workstation, cellular phone, or security system. A device in this definition can also be a connecting device such as a router a switch, a modem that changes the form of data, and so on.

Network Criteria

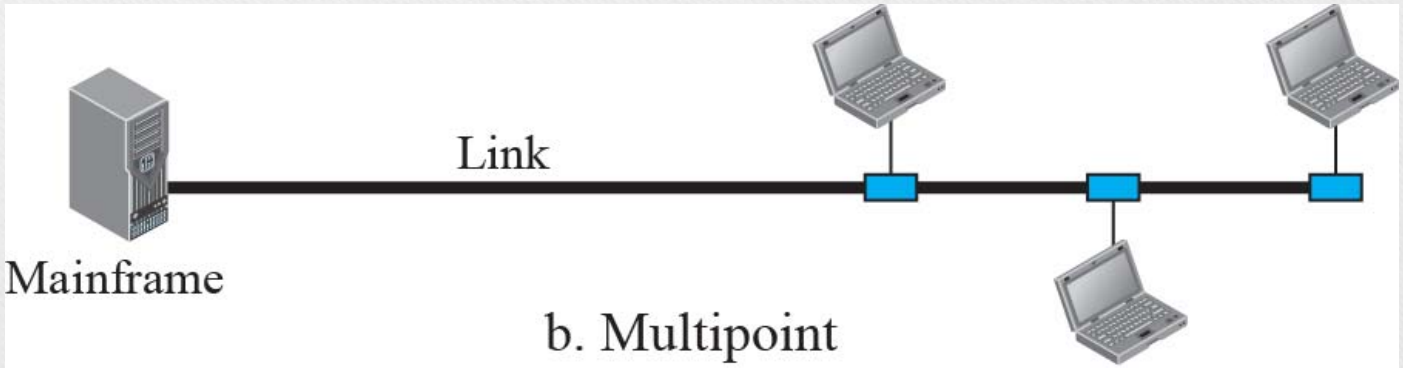
- Performance
  - Throughput
  - Delay
- Reliability
- Security

Physical Structures

- Type of Connection



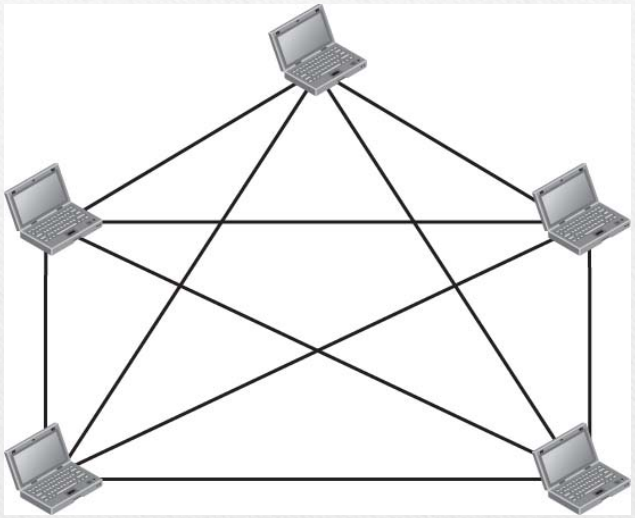
a. Point-to-point



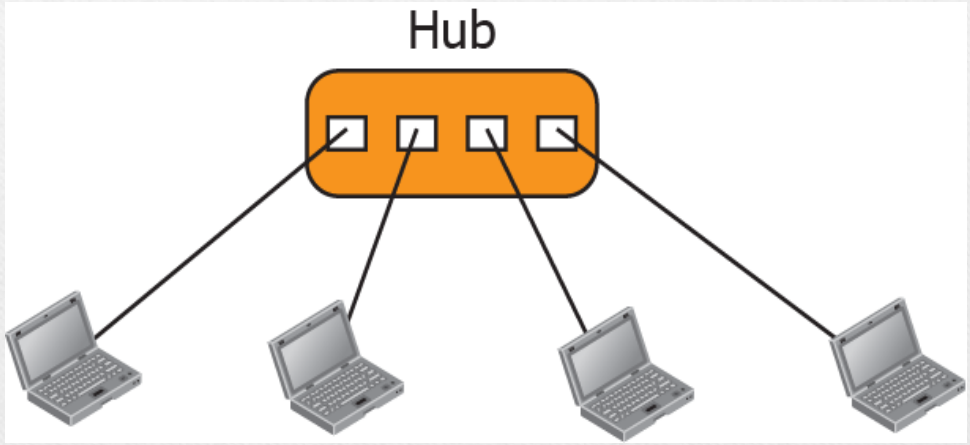
b. Multipoint

Physical Structures (Contd.)

- Physical Topology



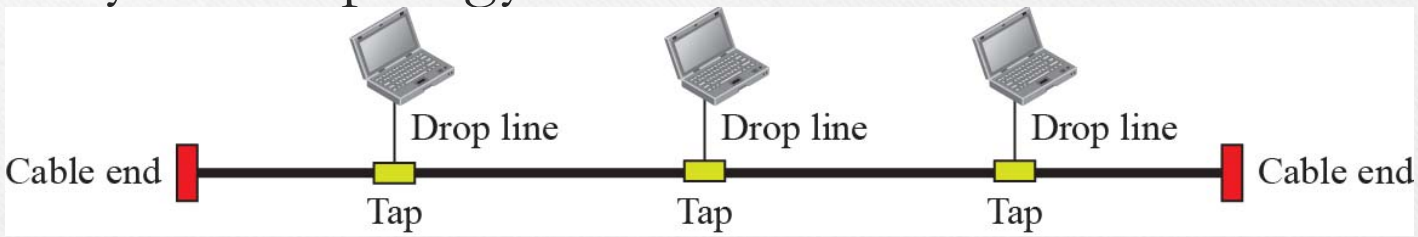
Mesh Topology



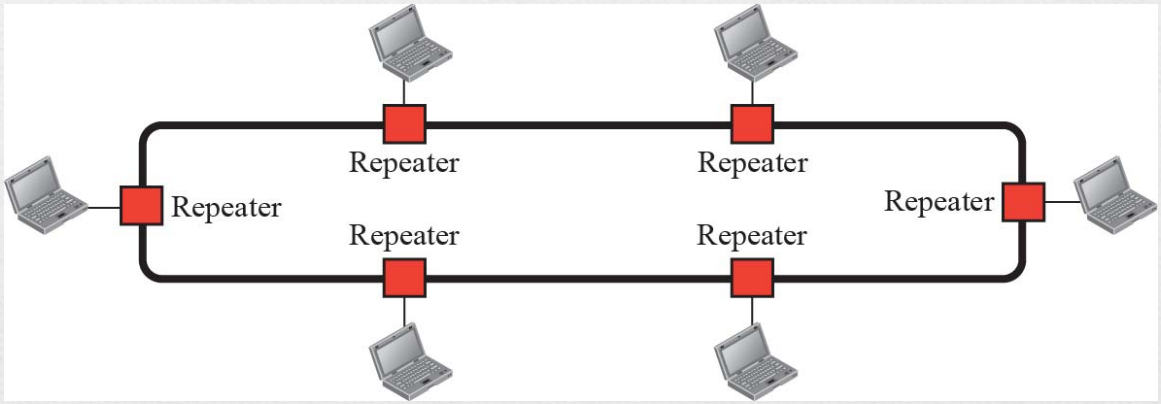
Star Topology

Physical Structures (Contd.)

- Physical Topology



Bus Topology



Ring Topology

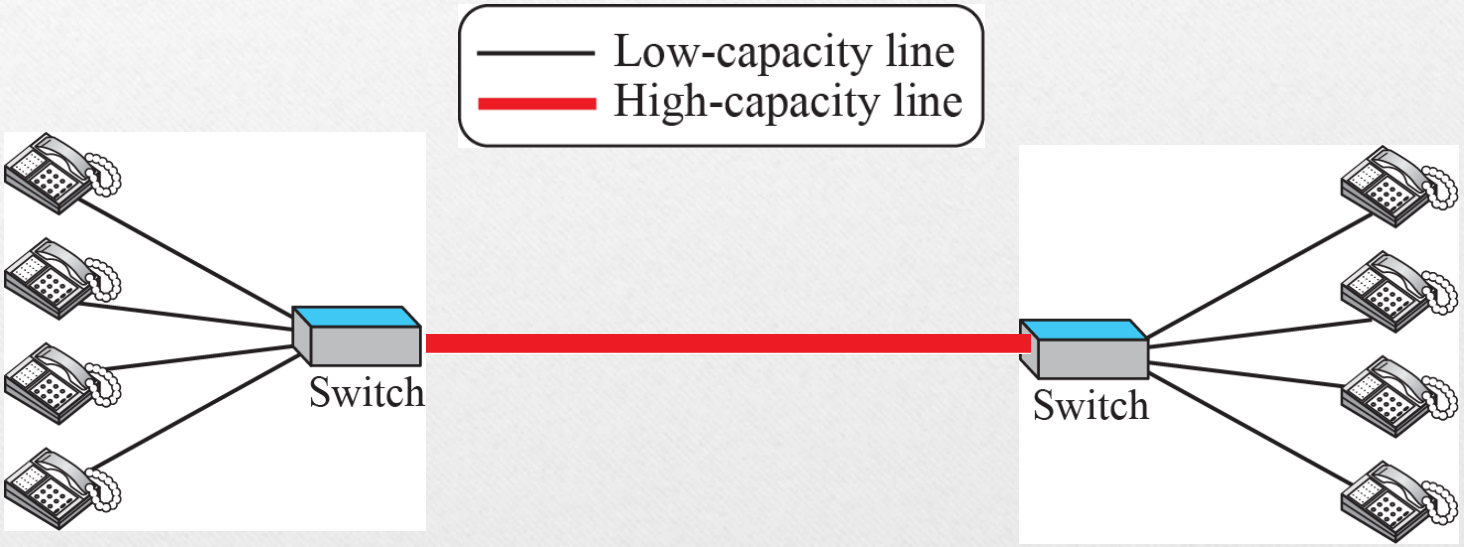
Network Types

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)
- Internetwork or Internet

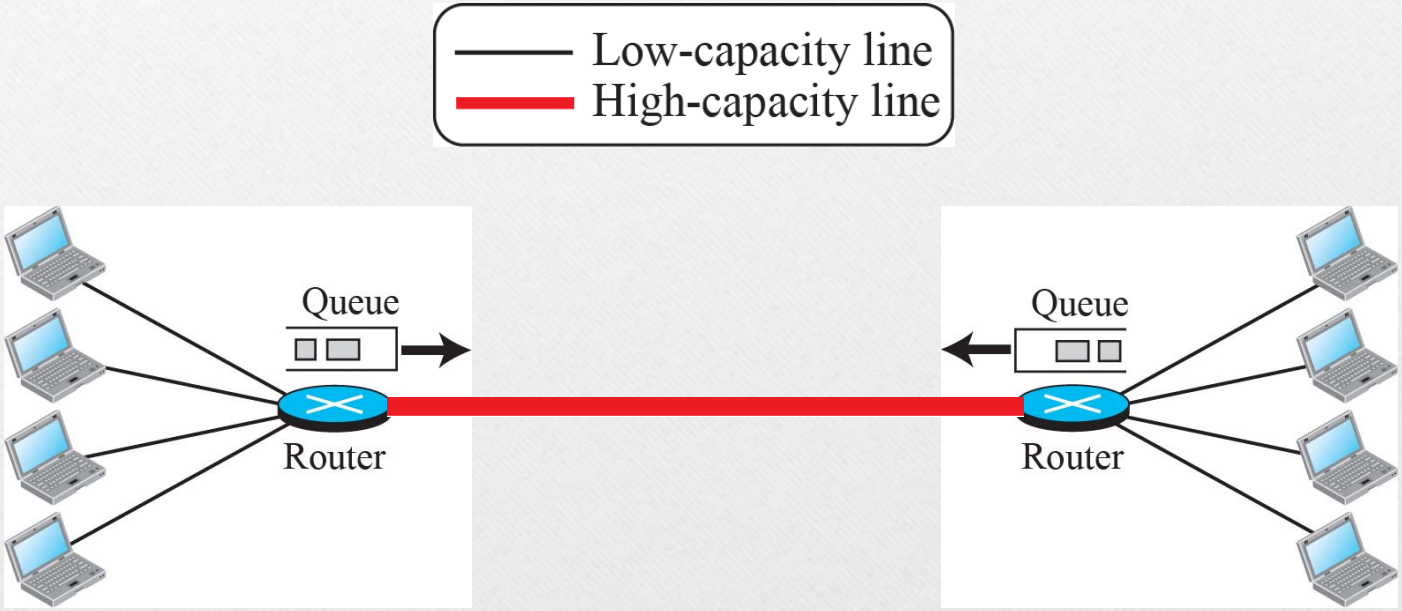
Switching

- An internet is a switched network in which a switch connects at least two links together. A switch needs to forward data from a network to another network when required. The two most common types of switched networks are circuit-switched and packet-switched networks.

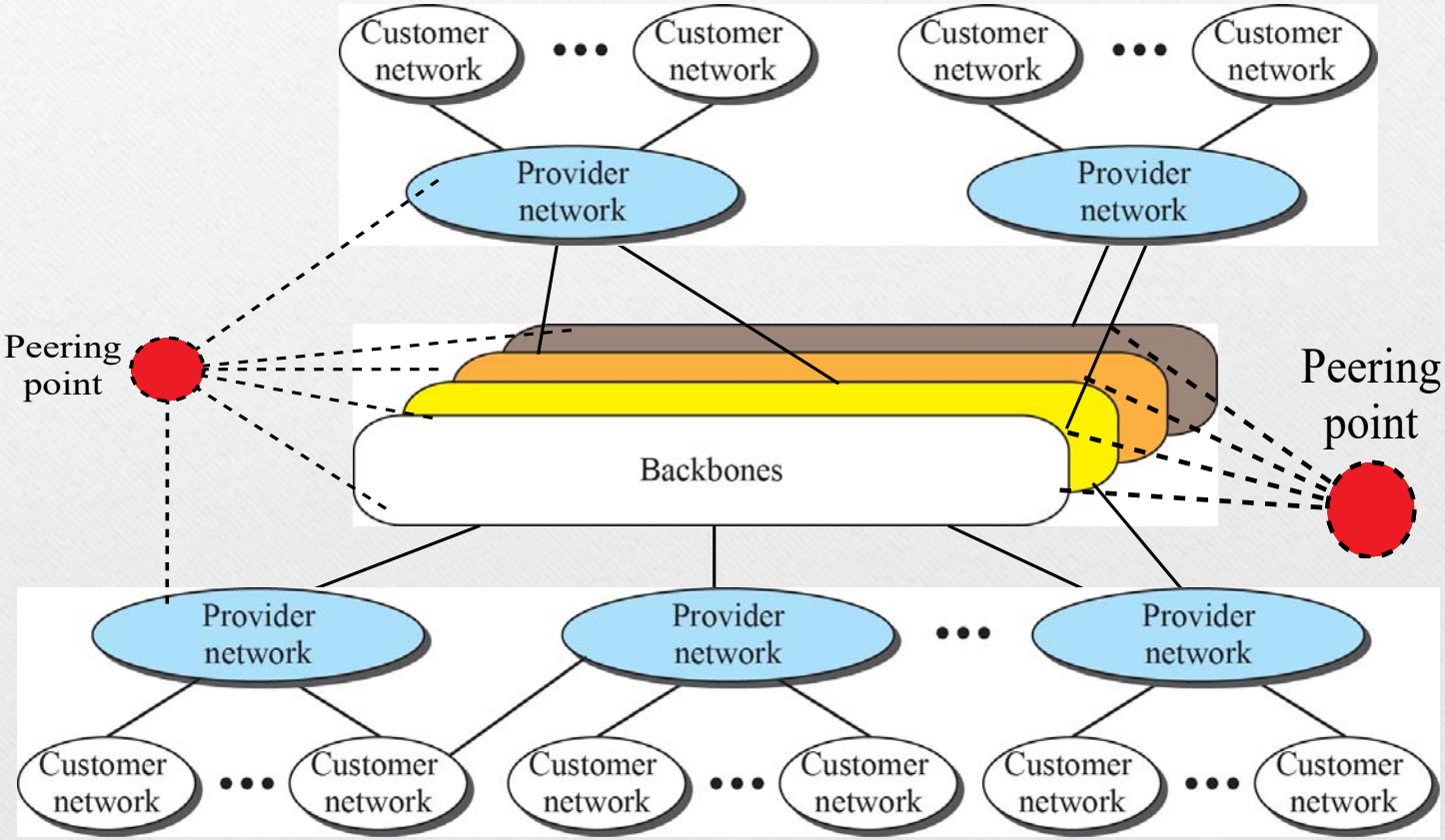
Circuit-Switched Network



Packet-Switched Network



The Internet

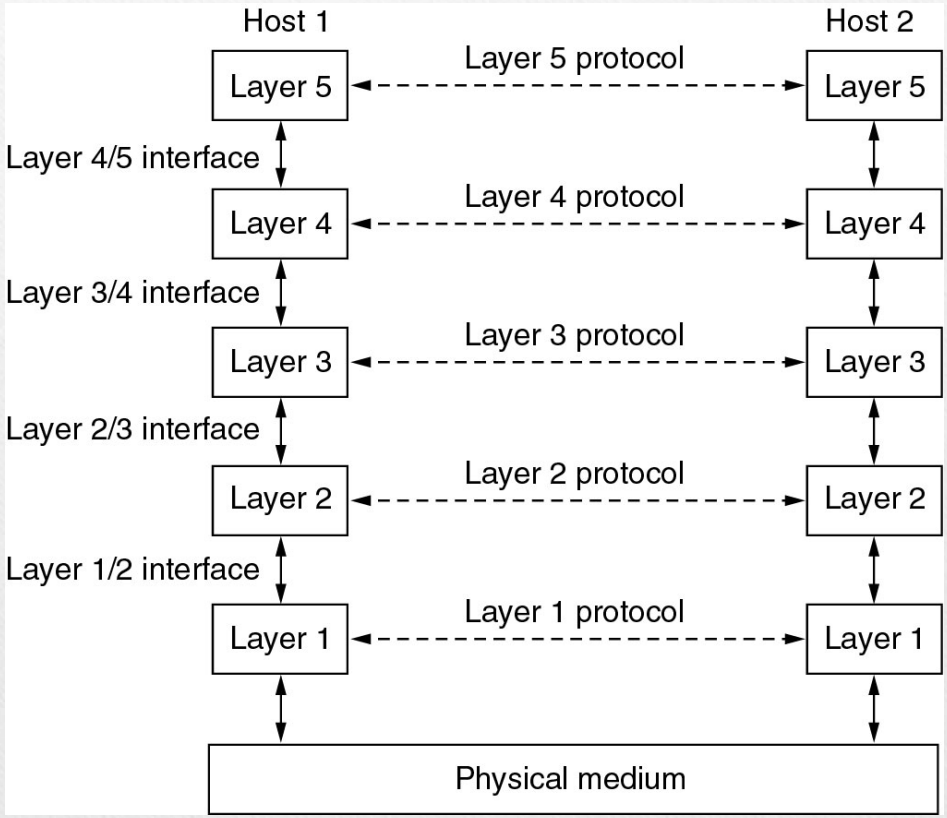


Network Models

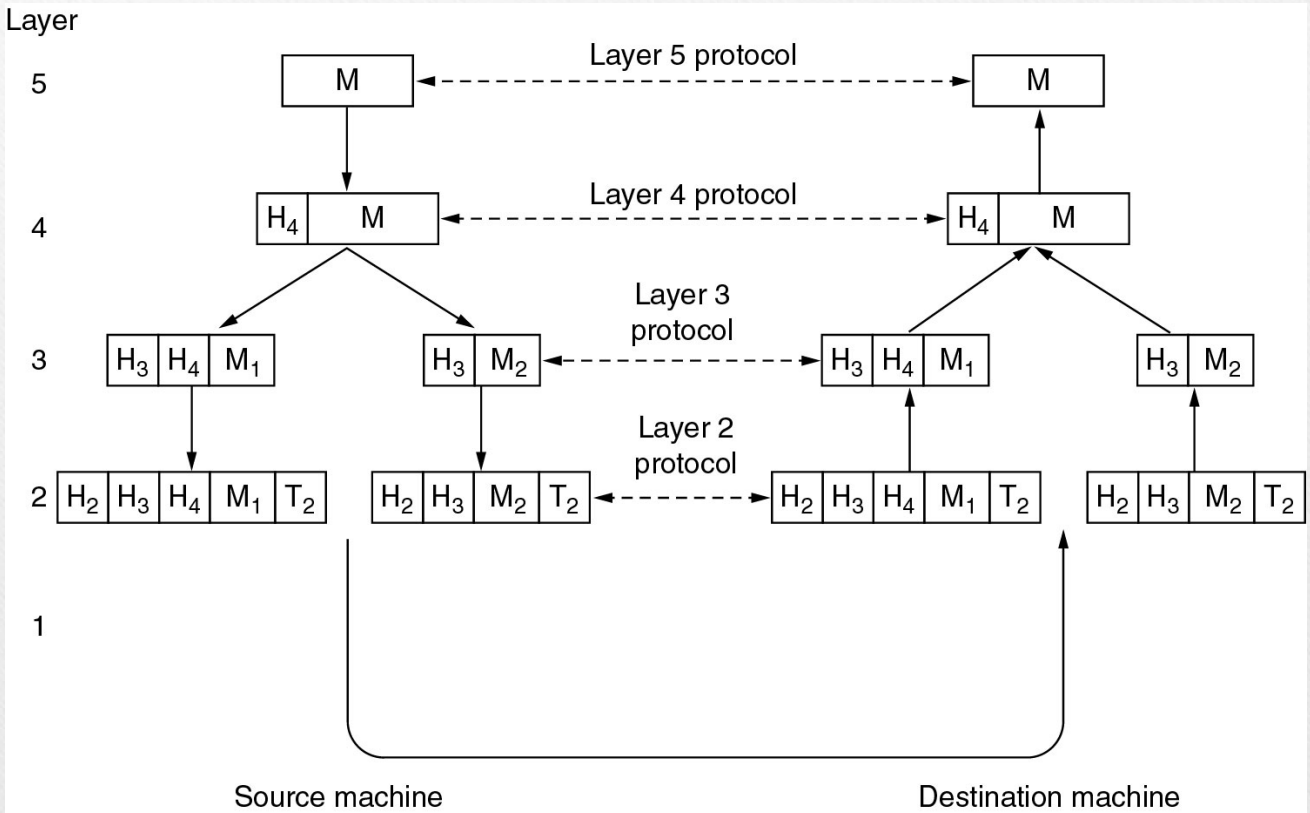
Protocols

- Basically, a protocol is an **agreement** between the communicating **peers** on how communication is to proceed.

Layering Principle



Layering Principle (Contd.)



Connection-Oriented vs. Connectionless

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

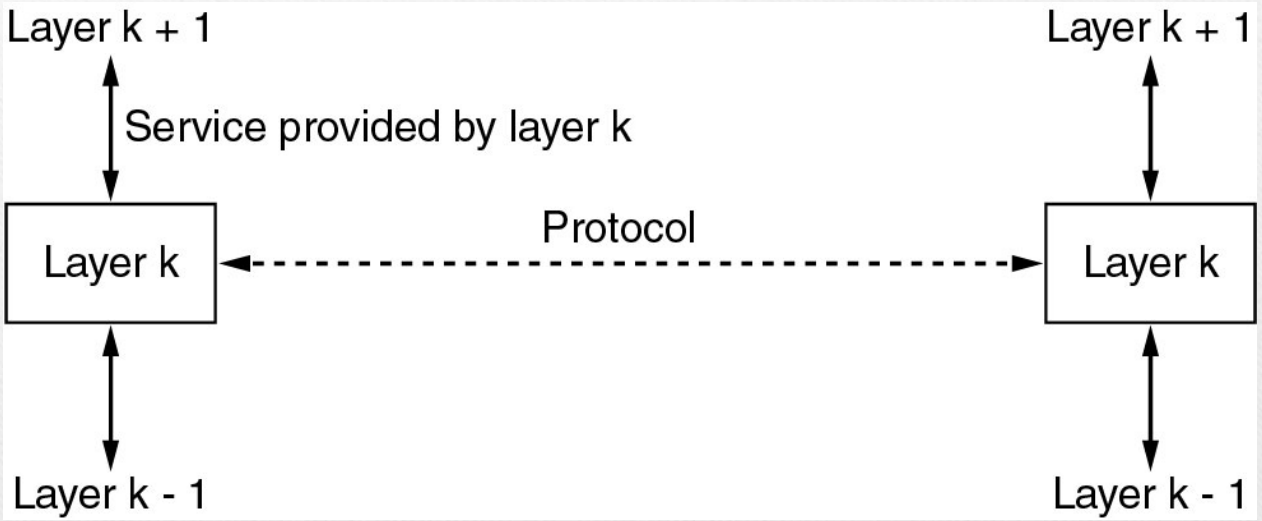
Service Primitives

- A service is formally specified by a set of primitives (basic operations) available to a user or other entity to access the service.

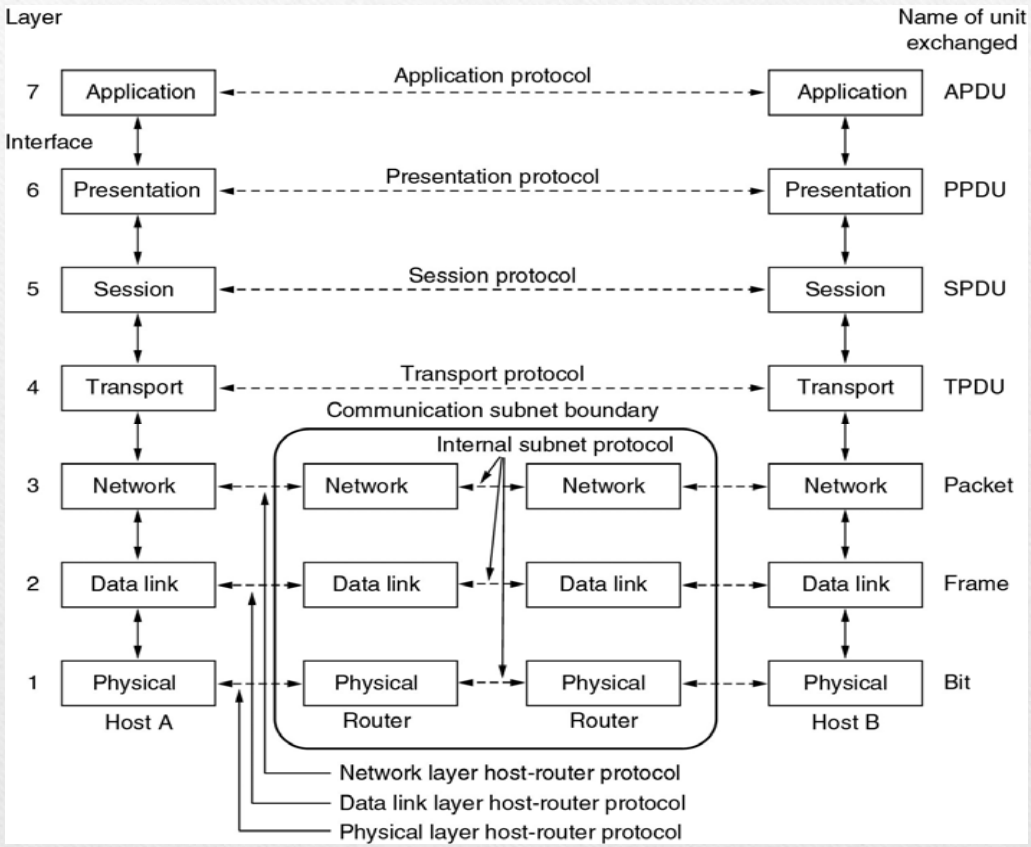
Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Service primitives for implementing a simple connection-oriented service

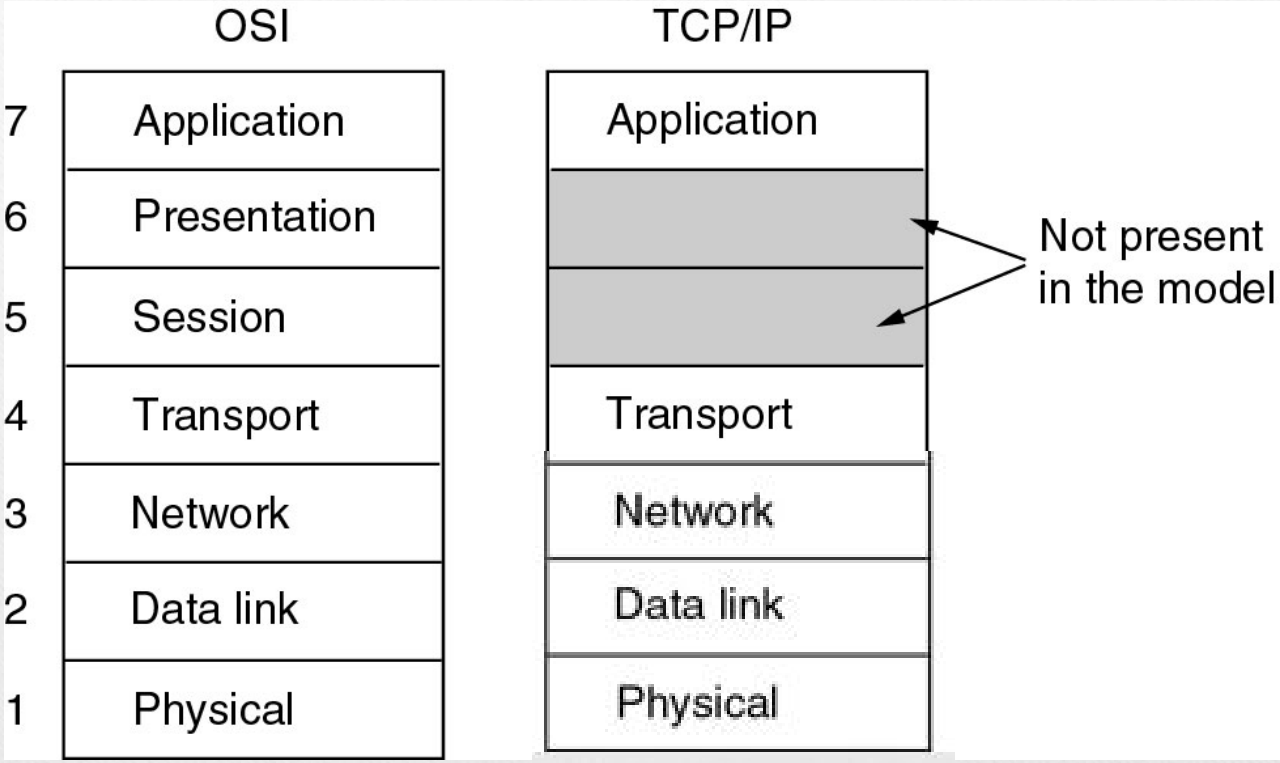
The relationship of Service



OSI Reference Model



TCP/IP Reference Model



Next Class

- OSI and TCP/IP Models (in detail)

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